3. There is no “market index” for illiquid asset classes.

   No investor receives the returns on illiquid indexes. An asset owner *never* receives the NCREIF return on a real estate portfolio, for example. The same is true for most hedge fund indexes and private equity indexes. In liquid public markets, large investors can receive index market returns and pay close to zero in fees. In contrast, NCREIF is not investable as it is impossible to buy all the underlying properties in that index. Since all asset owners own considerably fewer properties than the thousands included in NCREIF, they face far more idiosyncratic risk. While this large amount of idiosyncratic risk can boost returns in some cases, it can also lead to the opposite result. Returns to illiquid asset investing can be far below a reported index.
4. You cannot separate factor risk from manager skill.

   Tradeable and cheap index funds in bond and stock markets allow investors to separate systematic returns (or factor returns; see chapter 7) from management prowess. In illiquid markets, no such separation is possible: *investing in illiquid markets is always a bet on management talent*. The agency issues in illiquid asset markets are first-order problems, and I discuss them in Part III of this book. Agency problems can, and often do, overwhelm any advantages that an illiquidity risk premium may bring.

Taking into account data biases, the evidence for higher average returns as asset classes become more illiquid is decidedly mixed, as Ang, Goetzmann, and Schaefer (2011) detail.[22] But while there do not seem to be significant illiquidity risk premiums *across* classes, there are large illiquidity risk premiums *within* asset classes.

4.2. ILLIQUIDITY RISK PREMIUMS WITHIN ASSET CLASSES

Within all the major asset classes, securities that are more illiquid have higher returns, on average, than their more liquid counterparts. These illiquidity premiums can be accessed by dynamic factor strategies which take long positions in illiquid assets and short positions in liquid ones. As illiquid assets become more liquid, or vice versa, the investor rebalances. (Chapter 14 discusses how to allocate to these and other factors.)

U.S. Treasuries

A well-known liquidity phenomenon in the U.S. Treasury market is the *on-the-run/off-the-run bond spread*. Newly auctioned Treasuries (which are “on the run”) are more liquid and have higher prices, and hence lower yields, than seasoned

[22] Nevertheless, there are common components in illiquidity conditions across asset classes: when U.S. Treasury bond markets are illiquid, for example, many hedge funds tend to do poorly. See, for example, Hu, Pan, and Wang (2012).